

Daphne Qin

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EDUCATION

Rice University

Houston, TX

Bachelor of Science in Computer Science, GPA 3.94/4.0

Aug 2023 – May 2027

Relevant Coursework: Game Design, Data Structures and Algorithms, Discrete Mathematics, Computer Engineering, Computer Organization, Neuroengineering, Honors Linear Algebra

EXPERIENCE

Computational Biology Research Assistant

Jun 2024 – Present

Baylor College of Medicine

Houston, TX

- Design and implement a multi-stage data processing pipeline using Python and Shell Script used by 10 professors and researchers to process millions of lines of data.
- Optimize pipeline runtime by executing independent stages concurrently, reducing end-to-end runtime.
- Add validation and logging to support repeated execution and debugging across datasets.
- Collaborate in weekly sprint-style lab meetings using Agile workflows and present results to 100+ technical and non-technical audiences.

Machine Learning Research Assistant

Jan 2024 – Present

Baylor College of Medicine

Houston, TX

- Develop a semi-supervised machine learning algorithm using Python scikit-learn improving classification efficiency by 100% in brain structure characterization.
- Process electrical stimulation data using Python from over 50 epilepsy patients to analyze brain connectivity.

Teaching Assistant

Aug 2024 – May 2025

Rice University

Houston, TX

- Led a team of 50 TAs to organize an interactive, campus-wide treasure hunt to demonstrate the A-Star algorithm to students.
- Mentored over 300 students during office hours on concepts like Markov chains, graph search algorithms, and predictive model fitting.

Software Developer

Jan 2024 – Apr 2024

American Medical Women's Association

Houston, TX

- Implemented full-stack features using HTML/CSS/JavaScript to display personalized healthcare options.
- Led the creation of a patient-tailored questionnaire by collaborating with a panel of 5 physicians.
- Reached college students nationwide through Fox 26 Houston.

PROJECTS

AI-Driven Narrative Game | Godot, Google AI Studio

Nov 2025 – Present

- Design and implement a modular game system integrating AI-generated character dialogue with player-driven narrative state using Godot and custom Google AI integration.
- Build persistent dialogue memory, state tracking, and rule-based safety constraints to ensure deterministic, coherent character behavior across gameplay sessions.
- Engineer a low-latency interaction pipeline separating game logic from AI services, improving responsiveness and maintainability.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQLite, JavaScript, TypeScript, HTML/CSS, Shell

Frameworks: RESTful API, Flask, JUnit, Cypress

Developer Tools: Git, Google AI Studio, VS Code, Visual Studio, PyCharm, IntelliJ, Godot

Libraries: Pandas, NumPy, Matplotlib, scikit-learn